

5 Lab 4: Lap Time Simulation Pt.2

5.1 Introduction

The primary objective of this laboratory is to understand how to improve the overall racing car performance around the **Monza circuit** through vehicle setup optimization and strategic race management.

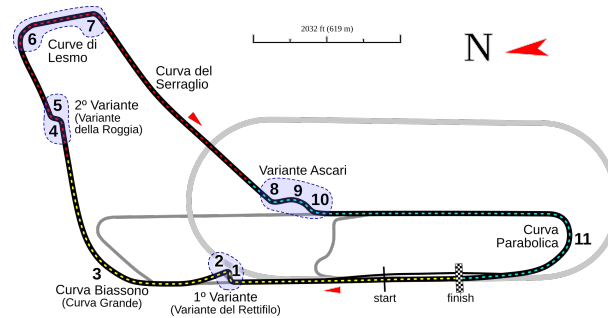


Figure 80

The analysis is conducted using a modified version of the OpenLap simulation software and focuses on two main topics: aerodynamic setup via **ride height adjustments** and energy saving through **lift and coast techniques**.

5.2 Exercise 1: Ride Heights Optimal Setup

5.2.1 Ground Effect and Aero Mapping

The underfloor of a racing vehicle generates a significant percentage of the car's total downforce. This phenomenon, known as "ground effect," is governed by Bernoulli's principle: as air flows under the floor, it accelerates, resulting in a drop in static pressure. This low-pressure area effectively "sucks" the car toward the ground.

While cars equipped with Venturi tunnels guide and accelerate the airflow before expanding it in the rear diffuser, a similar effect occurs with a flat floor.

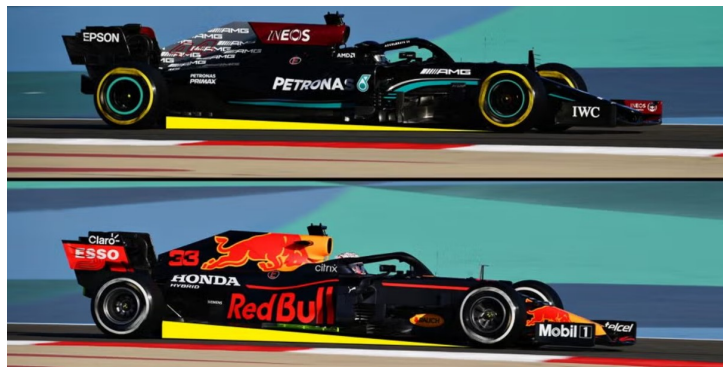


Figure 81

Increasing the inclination of the floor by keeping the front low and raising the rear (positive rake angle) amplifies the diffuser effect underneath the car. However, this must be

carefully managed to avoid aerodynamic stall if the angle becomes too steep or if the front ride height drops too low, choking the airflow.

To characterize the vehicle's aerodynamic behavior across all possible setup configurations, engineers rely on **"Aero Maps"**. These maps describe how the values of C_l , C_d , and aerodynamic Balance vary as a function of different FRH and RRH combinations.

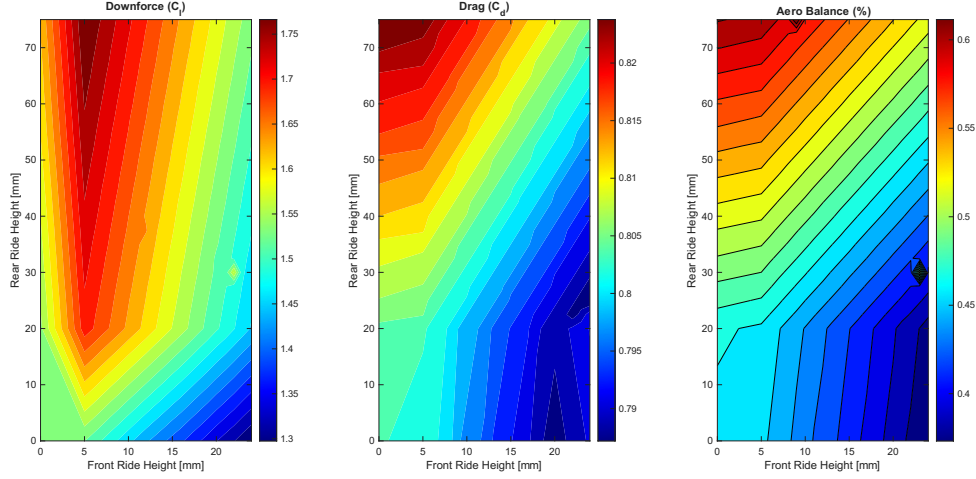


Figure 82: C_l , C_d and Balance maps

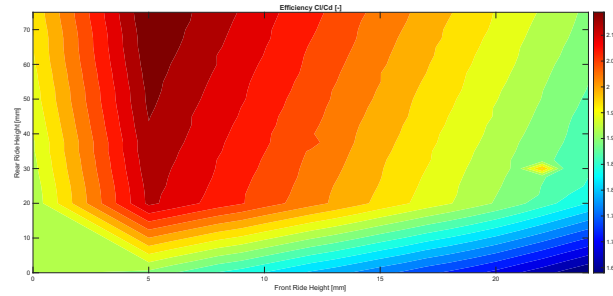


Figure 83: Efficiency (C_l/C_d) Map

5.2.2 Dynamic Aerodynamics Evaluation

The static ride heights set in the garage (FRH_{static} and RRH_{static}) are not maintained when the vehicle is moving. As speed increases, the generated downforce compresses the front and rear suspensions, dynamically lowering the car closer to the ground. Consequently, the aerodynamic coefficients (C_d , C_l) and the aero balance (Bal) change continuously with velocity.

To perform a reliable lap time simulation, it is necessary to map these aerodynamic parameters as a function of speed. This is achieved through a quasi-static iterative approach:

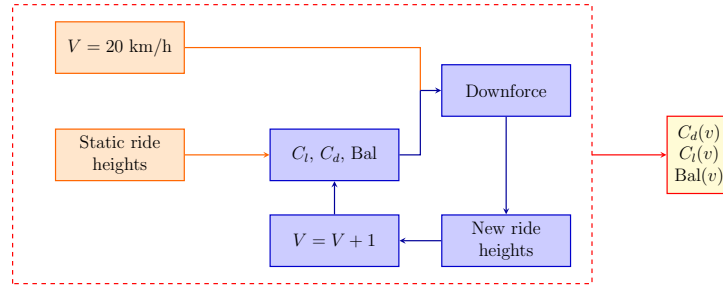


Figure 84

At each velocity step, the current ride heights are used to interpolate the aerodynamic coefficients from the Aero Maps. These coefficients are then used to calculate the new downforce, which determines the new compressed ride heights based on the suspension stiffness (K_{zf} and K_{zr})

```

1 %% Static (Garage) Setup:
2 FRH_static = ; % [0, 24] mm
3 RRH_static = ; % [0, 75] mm
4 Kzf = ; % N/mm [206, 264, 294, 324];
5 Kzr = ; % N/mm
6 veh = load('OpenVEHICLE Vehicles\OpenVEHICLE_Tatuus.mat');
7 Area = veh.A;
8 %% Cl(V), Cd(V) and Bal(V)
9 V_vec = 0:230; %km/h
10 rho_air = 1.225; % kg/m^3
11 FRH_current = FRH_static;
12 RRH_current = RRH_static;
13
14 Cl_vec = zeros(1, length(V_vec));
15 Cd_vec = zeros(1, length(V_vec));
16 Bal_vec = zeros(1, length(V_vec));
17 FRH_vec = zeros(1, length(V_vec));
18 RRH_vec = zeros(1, length(V_vec));
19
20 for ii = 1:length(V_vec)
21 Cd_current = interp2(RRH, FRH, cd, RRH_current, FRH_current, "spline");
22 Cl_current = interp2(RRH, FRH, cl, RRH_current, FRH_current, "spline");
23 Bal_current = interp2(RRH, FRH, Bal, RRH_current, FRH_current, "spline");
24
25 Cd_vec(ii) = Cd_current;
26 Cl_vec(ii) = Cl_current;
27 Bal_vec(ii) = Bal_current;
28
29 DownForce = 0.5*rho_air*Cl_current*veh.A*(V_vec(ii)/3.6)^2;
30
31 FRH_current = FRH_static - (Bal_current*DownForce)/Kzf;
32 RRH_current = RRH_static - ((1 - Bal_current)*DownForce)/Kzr;
33 FRH_vec(ii) = FRH_current;
34 RRH_vec(ii) = RRH_current;
35
36 if FRH_current <= 0 || RRH_current <= 0
37     disp(['Error: Bottoming at ', num2str(V_vec(ii)), ' km/h]);
38     break;
39 end
40 end
  
```

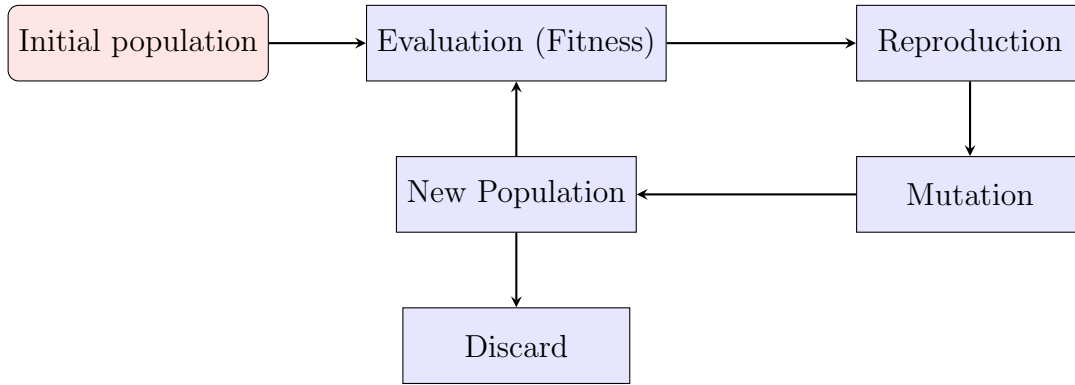
To prevent any bottoming, a safety constraint aborts the simulation if $FRH \leq 0$ or $RRH \leq 0$

The $C_l(V)$, $C_d(V)$ and $Bal(V)$ profiles enter the simulation script `OpenLAP` by means of interpolation functions:

```
1 cl_fun = @(v) interp1(V_vec, Cl_vec, v, 'linear', 'extrap');
2 cd_fun = @(v) interp1(V_vec, Cd_vec, v, 'linear', 'extrap');
3 Bal_fun = @(v) interp1(V_vec, Bal_vec, v, 'linear', 'extrap');
4
5 save("functions", "Bal_fun", "cd_fun", "cl_fun");
```

5.2.3 Genetic Algorithm Optimization for Vehicle Setup

To identify the optimal setup for the Monza layout, a **Genetic Algorithm (GA)** is implemented.



Fitness Function Formulation Each setup needs to be evaluated in terms of its performance along the race track. To define a representative fitness function, the influence of lift and drag variations on the Monza lap time is quantified by means of an aerodynamic sensitivity analysis derived from a pre-calculated Isochrones map.

A Multiple Linear Regression (Least Squares Method) was performed on a specific, useful subset of the C_l and C_d operating window. The regression yielded the following sensitivity coefficients:

- $c_1 = 2.4798$ (Sensitivity to C_l)
- $c_2 = 16.4294$ (Sensitivity to C_d)

These coefficients physically demonstrate the nature of the Monza circuit: the drag penalty is significantly more impactful than the downforce gain. These extracted weights are then adopted as the core of the GA's fitness function:

$$F = c_1 \cdot C_{l,mean} - c_2 \cdot C_{d,mean}$$

where $C_{l,mean}$ and $C_{d,mean}$ represent the mean Lift and Drag Coefficients. Crucially, these mean values are not evaluated over the entire speed range (0-230 km/h), but are calculated strictly for speeds exceeding 130 km/h:

```

1 Cl_mean = mean(Cl_vec(V_vec > 130));
2 Cd_mean = mean(Cd_vec(V_vec > 130));

```

This velocity threshold is intentionally applied to isolate the vehicle’s aerodynamic efficiency within the high-speed operating regime. Since Monza is a remarkably fast circuit, low-speed aerodynamic data is largely irrelevant to the overall lap time. Filtering the data ensures the algorithm optimizes the setup for the track’s dominant characteristic.

Finally, a severe penalty ($F = -1000$) is added whenever the dynamic ride heights (FRH or RRH) drop to or below 0 mm at any point during the simulation. In such cases, the setup is considered invalid.

Chromosome Architecture and Mixed Coding The optimization requires handling a mixed design space:

- **Continuous Variables (Ride Heights):** To find the absolute **theoretical optimum** of the aerodynamic maps, the ride heights require a mathematical resolution of at least 0.01 mm. A proper set of bits are adopted and allocated for each of them to ensure a proper resolution based on their range of variability.

Variable	Allocated Bits	Levels (2^n)	Range [min, max]	Resolution
Front Ride Height (FRH)	11	2048	[0, 24] mm	0.0117 mm
Rear Ride Height (RRH)	13	8192	[0, 75] mm	0.0092 mm

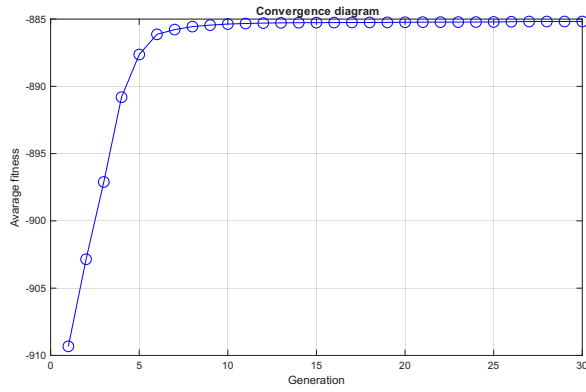
The continuous variables are first mapped to integer decimal values using a standard linear quantization formula:

$$val_{int} = \text{round} \left(\frac{x - x_{min}}{x_{max} - x_{min}} \cdot (2^{n_{bits}} - 1) \right)$$

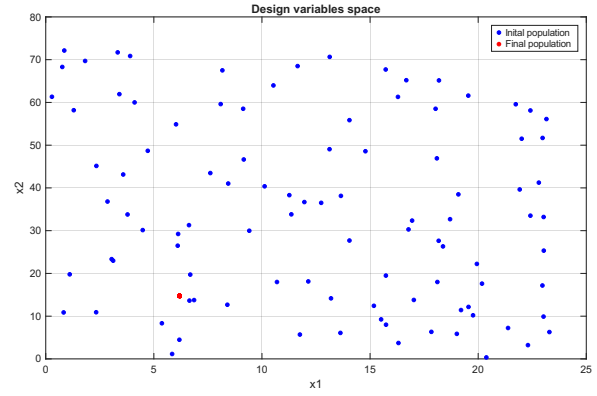
The resulting integer val_{int} is then directly converted into its corresponding binary string representation.

- **Discrete Variables (Suspension Stiffness):** To avoid generating non-existent spring stiffnesses, which must be selected from an available catalog ($K = [206, 264, 294, 324]$ N/mm), an index-based mapping strategy is implemented. The 2 bits allocated for each spring are used to generate a 0-based index (00, 01, 10, 11) that points directly to the catalog array during the decoding phase.

5.2.4 Optimization Outcomes



(a) Fitness convergence diagram.



(b) Population Evolution

Variable	Optimum (theoretical) value
FRH_{static}	6.61 mm
RRH_{static}	14.97 mm
K_{zf}	324.0 N/mm
K_{zr}	324.0 N/mm

Predicted LapTime	110.970 sec
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The dynamic operating window (**black dotted line**) of the optimal setup is plotted over the raw Aero Maps.

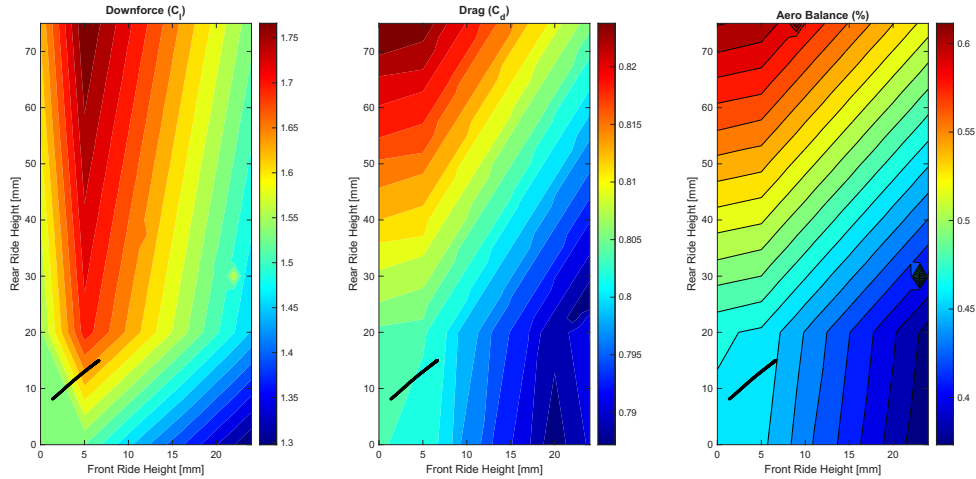


Figure 86

Although the sensitivity analysis shows that drag reduction is crucial at Monza, the algorithm successfully identifies an optimal compromise between aerodynamic resistance and downforce.

A layout focused exclusively on minimum drag would lead to extreme speeds on the straights but would heavily penalize the vehicle through Monza's demanding braking zones

and corners. To balance this, the algorithm exploits the aero-mechanical compliance of the suspension system:

At lower speeds, which are typical of cornering conditions, the aerodynamic forces are relatively small, meaning the suspension experiences minimal compression. Consequently, the vehicle operates close to its static setup heights, where the floor maintains a higher rake angle and remains inside a high-downforce region ($C_l \approx 1.65 - 1.7$) to maximize lateral grip.

As the vehicle accelerates down the straights, the increasing downforce compresses the springs, forcing both the *FRH* and *RRH* to drop. This mechanical "squatting" effectively flattens the car and brings the operational point out of the high-lift zone and into a lower drag region ($C_d < 0.795$).

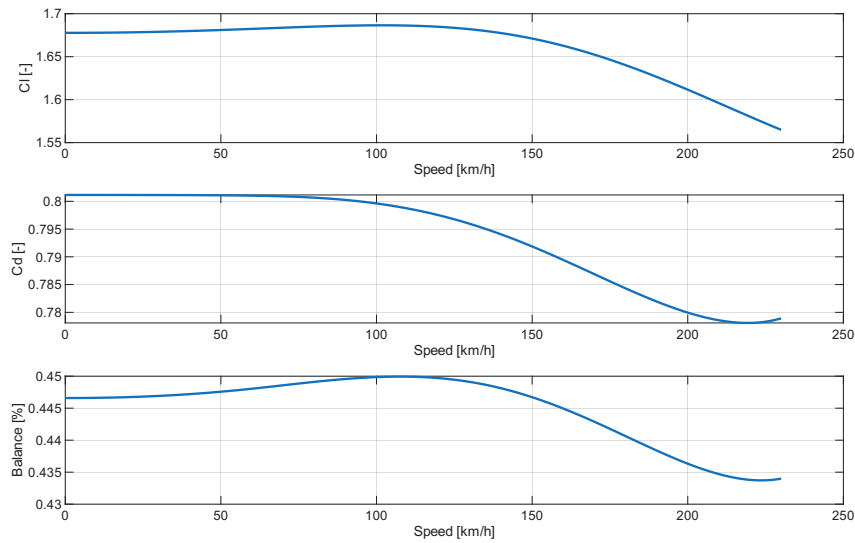


Figure 87

It is crucial to notice that while the absolute aerodynamic forces (Downforce and Drag) increase quadratically with vehicle speed, the aerodynamic coefficients (C_l and C_d) exhibit a decreasing trend along the simulated speed.

This behavior comes from the dynamic compression of the suspension system: as the vehicle accelerates, the increasing downforce compresses the car towards the ground, causing two main aerodynamic effects:

1. **Loss of Rake:** The reduction of the Rear Ride Height decreases the overall rake angle of the vehicle. A flatter floor reduces the expansion volume of the rear diffuser, leading to a drop in the downforce coefficient (C_l). Simultaneously, the reduced frontal projection and lower induced drag contribute to a beneficial drop in the drag coefficient (C_d).

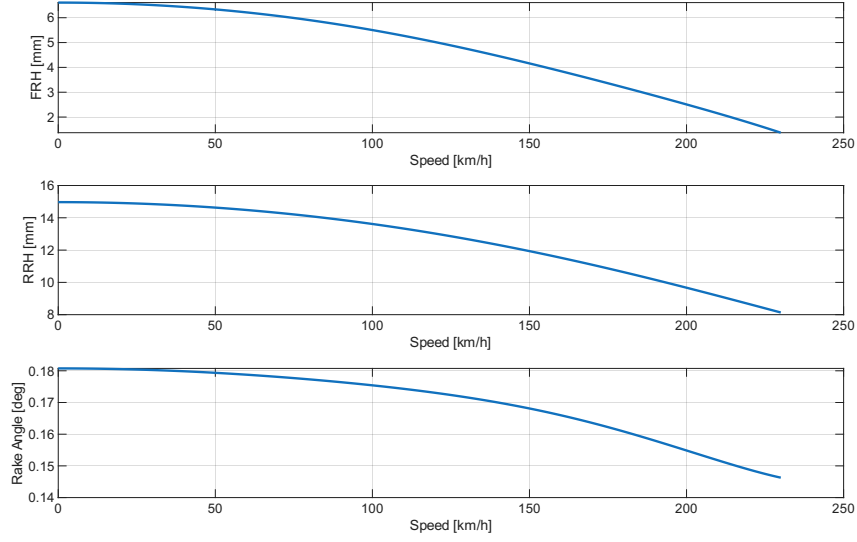


Figure 88

2. **Aerodynamic Choking:** As the Front Ride Height approaches 0 mm at maximum velocity, the cross-sectional area available for the airflow to enter the underbody becomes critically narrow. This chokes the mass flow rate underneath the car, further reducing the floor's capability to generate low pressure, thus decreasing the overall C_l .

This phenomena highlight the importance of selecting stiff suspension springs (as correctly identified by the GA) to prevent severe choking and maintain a stable aerodynamic platform at the end of the straights.

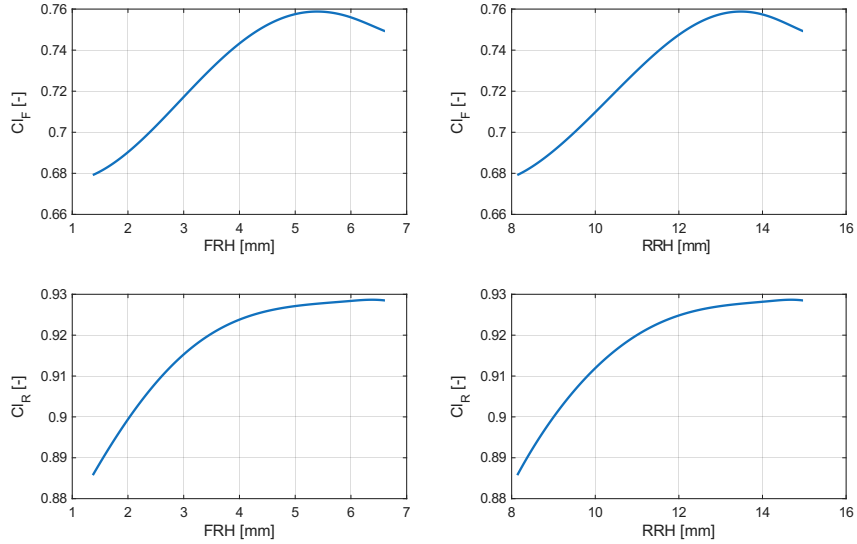


Figure 89

5.2.5 Theoretical vs. Practical Application

Implementation of a Safety Ride Height Margin To align the optimization with realistic trackside practices, a final run of the Genetic Algorithm is performed by intro-

ducing practical **minimum clearance thresholds**: $FRH_{safe} = 5.0$ mm and $RRH_{safe} = 10.0$ mm.

In real-world racing, maintaining a strict safety margin from the ground is essential to protect the vehicle and ensure stability against several operational factors:

- **Track Irregularities:** Prevents the chassis from hitting localized asphalt bumps (*bottoming*), which would momentarily unload the tires and cause a sudden loss of mechanical grip.
- **Kerb Riding:** It allows the driver to aggressively attack the kerbs without the risk of damaging the underbody of the car.
- **Technical Legality:** It limits the friction wear of the underbody skid block (*plank*), ensuring the car complies with the maximum permissible wear defined by FIA technical regulations at the end of the race.

The GA's fitness function is modified to apply a penalty ($F = -1000$) as soon as the dynamic ride heights drop below these values at any speed. This constraint forces the evolutionary algorithm to converge toward a mechanically robust setup that inherently accounts for real-world trackside safety.

Variable	Feasible Setup value
FRH_{static}	10.85 mm
RRH_{static}	20.13 mm
K_{zf}	324.0 N/mm
K_{zr}	324.0 N/mm

Predicted LapTime	111.323 sec
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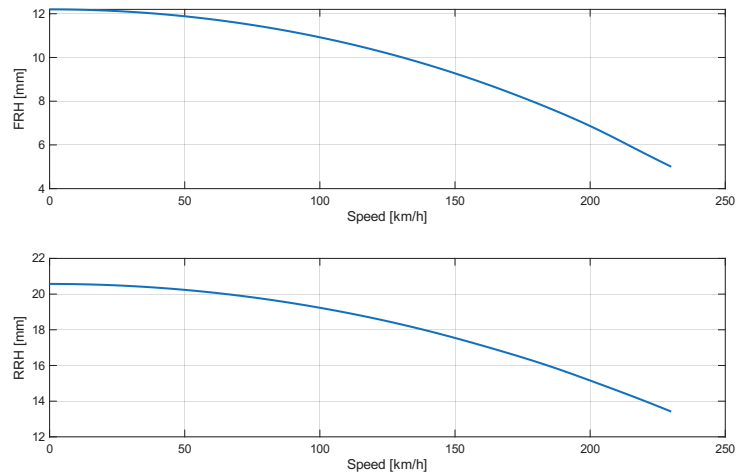


Figure 90

Heights Adjustments In real track operations, mechanics adjust ride heights by adding small spacers. These physical adjustments typically allows a limited resolution (approximately 0.5 mm).

Consequently, the RH values are rounded to the nearest feasible mechanical increment.

Variable	Feasible Setup value
FRH_{static}	11.00 mm
RRH_{static}	21.00 mm
K_{zf}	324.0 N/mm
K_{zr}	324.0 N/mm

Predicted LapTime	111.330 sec
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This minor approximation does not significantly alter the aerodynamic platform, ensuring that the actual vehicle performance closely matches the simulated feasible predictions while remaining physically adjustable in the pit lane.

5.2.6 Model Limitations

While the implemented quasi-static model provides an effective optimization tool for lap time simulation, it relies on several Aerodynamic simplifications that neglect complex physical phenomena occurring in real track conditions. The primary limitations include:

- **Neglection of Roll and Steering Effects:** The provided Aero Maps only evaluate symmetrical vertical displacements (FRH and RRH). During cornering, lateral load transfer induces vehicle roll, creating asymmetrical ride heights between the inner and outer sides of the floor. Furthermore, front wheel steering angles distort the airflow directed under the chassis. Both phenomena significantly alter C_l , C_d , and the aerodynamic balance in corners.
- **Extreme Proximity to Ground and Aerodynamic Choking:** The model allows the vehicle to run with a very low Front Ride Height without any severe consequences. In reality, such extreme proximity to the track surface would induce an immediate and severe aerodynamic choking. Additionally, real-world track roughness and bumps would cause the chassis to strike the ground (*bottoming out*), destabilizing the mechanical platform—a risk that this idealized, smooth-surface simulation completely neglects.

In conclusion, the results obtained are purely theoretical and serve as a first approach for understanding the limits of the aerodynamic platform applied to a performance investigation. In real track operations, an engineer must select a more conservative and "safe" setup. Accounting for unpredictable variables—such as track bumps, transients and structural tire deflection—requires leaving a reasonable physical margin also for the ride heights.

5.3 Exercise 2: Race Strategy with Lift and Coast

The "Lift and Coast" (L&C) technique consists of releasing the throttle pedal several meters before the ideal braking point, allowing the vehicle to "coast" using its aerodynamic drag to decelerate before the friction brakes are applied. Beyond reducing fuel consumption, this strategy significantly mitigates thermal and mechanical stress on the braking system and on the tires, shifting the trade-off from single lap time maximization to global race distance optimization.

5.3.1 L&C Sensitivity on LapTime

To mathematically understand why L&C is effective, we must analyze how localized speed variations affect the total lap time.

The total lap time T is the integral of the inverse of velocity over the circuit length L :

$$T = \int_0^L \frac{1}{v} ds$$

If a perturbation Δv is introduced (e.g., due to coasting), the resulting variation in lap time ΔT can be linearized through a first-order Taylor expansion:

$$\Delta T \approx \int_0^L \left(-\frac{\Delta v}{v^2} \right) ds$$

This fundamental relationship shows that the lap time sensitivity to speed losses is inversely proportional to the square of the local velocity ($1/v^2$). Consequently, a loss of speed (Δv) at very high velocities (such as at the end of a long straight) yields an extremely small time penalty ($\Delta T \rightarrow 0$). Conversely, dropping speed in low-speed corners or during acceleration phases drastically increases the lap time.

To have a practical feedback of the analytical relationship, it is possible to visualize the sensitivity index for each braking zone on the Monza circuit.

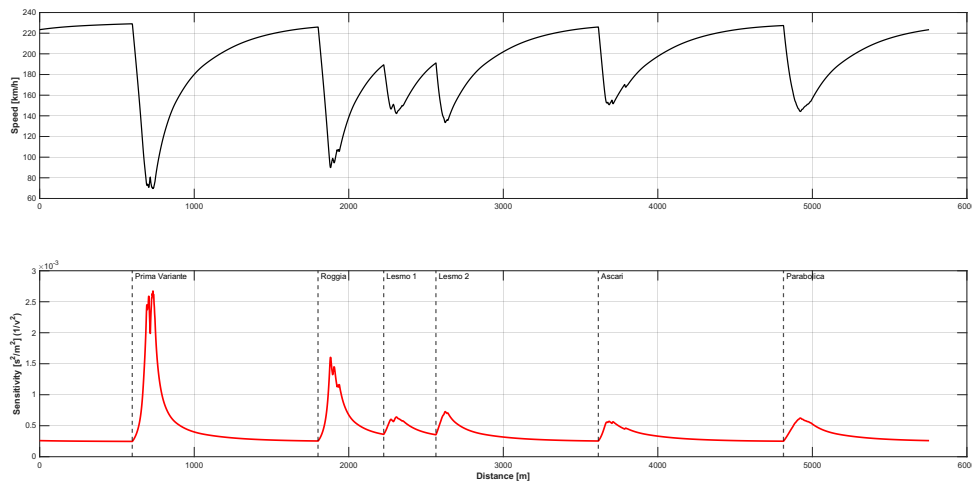


Figure 91

A theoretical efficiency ranking for Lift and Coast deployment across Monza's main corners can be established:

Corners (Best to Worse)	Approaching Top Speed [km/h]	Sensitivity $1/v^2$ [s^2/m^2]
Prima Variante	229.11	2.469e-04
Parabolica	227.44	2.505e-04
Ascari	226.01	2.537e-04
Seconda Variante (Roggia)	225.95	2.538e-04
Lesmo 2	191.20	3.545e-04
Lesmo 1	189.41	3.612e-04

The laboratory task specifically restricts the optimization scope to the distinction between **Prima Variante** (the most efficient zone) and **Lesmo 1** (the least efficient zone)

5.3.2 Prima Variante vs Lesmo 1

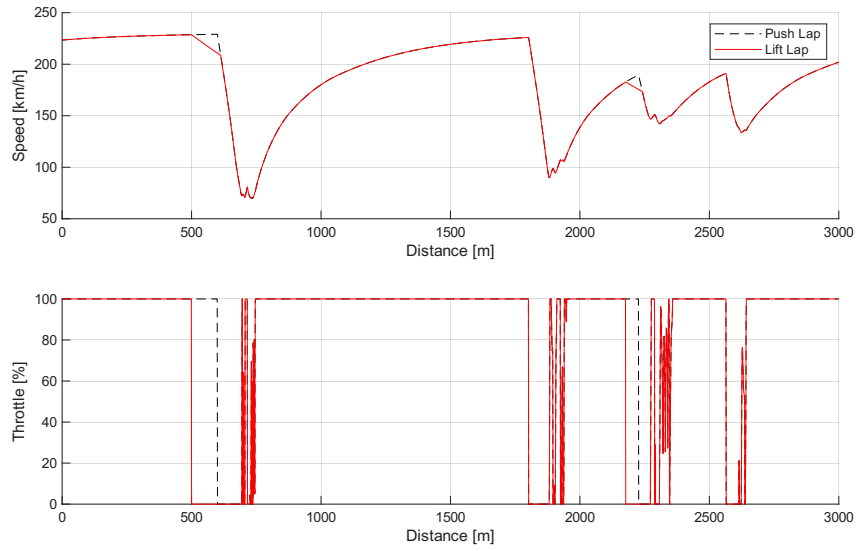


Figure 92

An initial spatial sensitivity analysis is conducted to evaluate the isolated effect of lift distance (m) on lap time penalty (ΔT_{lap}), fuel consumption savings (Δm_{fuel}), and brake energy reduction (ΔE_{brake}).

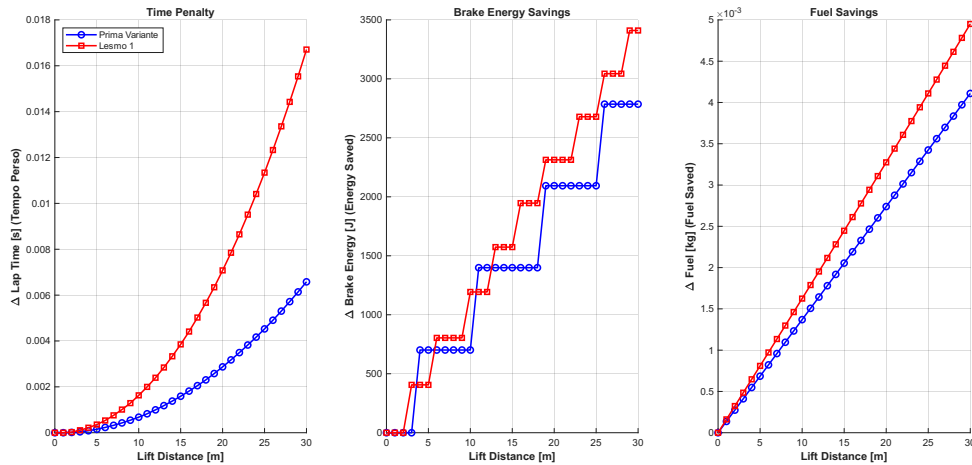


Figure 93

The ΔT_{lap} vs m confirms what previously said: coasting at lower speeds increases the overall time penalty.

However, Lift and Coast at Lesmo 1 might counterintuitively appears more advantageous for fuel and brake energy conservation. As observed in the spatial plots, for a given lift distance, the vehicle saves more absolute energy at Lesmo 1 than at Prima Variante. This occurs because the vehicle approaches Lesmo 1 at a lower speed; consequently, it spends more time covering the same coasting distance with a closed throttle/ released brakes.

To definitively answer which corner is more convenient for L&C deployment we can evaluate the Efficiency Index (IE), defined as the ratio of resource saved to time lost ($\Delta \text{Resource} / \Delta \text{Time}$).

$$IE_{fuel} = \frac{\Delta \text{Fuel Saved}}{\Delta \text{Lap Time}} \quad [\text{kg/s}]$$

$$IE_{brake} = \frac{\Delta \text{Brake Energy Saved}}{\Delta \text{Lap Time}} \quad [\text{J/s}]$$

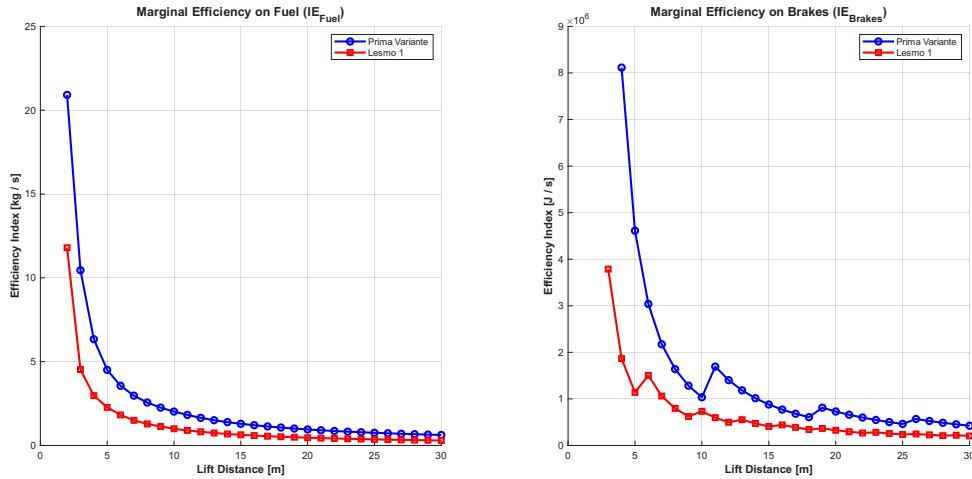


Figure 94

The Efficiency Indices are not constant values, but shows a decaying trend with the lift distance. The initial meters of lift brings enormous efficiency peaks, due to the low amount of lap time penalty; the more we lift before the braking zone, the higher is the energy content saved, but at a cost of a larger time penalty, which lowers the overall Efficiency Index value.

The Efficiency Indices prove once more that **Prima Variante is better than Lesmo 1 for Lift and Coast strategy**. Infact, in Lesmo 1, the car is forced to sacrifice a higher amount of lap time to achieve any meaningful savings.

5.3.3 40-Laps Race Optimal Strategy

The following chapter has the aim of answering the following question: is it better to start a race with a under-loaded car and perform Lift and Coast or load the car in order to push every single lap?

To prove the difference between the two cases, a series of different race strategies on a **40-laps Race** is computed in order to asses which is the faster one:

1. **Full Push Race:** No L&C strategy, the car is fully loaded to push every lap from start to finish
2. **Constant Lift Race:** L&C is performed every lap with a fixed lift distance
3. **Adaptive (Dynamic) Lift Race:** L&C is performed changing the lift distance depending on the current vehicle's mass and race lap

5.3.4 Push Race

A first simple strategy would be loading the car with the right amount of fuel necessary for completing the whole race without performing any L&C. The car is heavier at the start but the driver can push every lap without managing the fuel consumption.

To evaluate what is the right amount of fuel to load on the car, a **Predictor-Corrector Algorithm** is created in order to guess the exact fuel amount in order the vehicle to finish the race with the tank empty.

```

1 function [carburante] = Calcolo_Carburante(lift_prima_var,pos_lift_prima_var,lift_lesmo,
2     pos_lift_lesmo)
3 lift_prima_var = lift_prima_var;
4 pos_lift_prima_var = pos_lift_prima_var;
5 lift_lesmo = lift_lesmo;
6 pos_lift_lesmo = pos_lift_lesmo;
7 veh = load('OpenVEHICLE_Vehicles/OpenVEHICLE_Tatuus.mat') ;
8
9 M_base = veh.M;
10 LapTime = [];
11 fuel_cons = [];
12
13 m_incr = 0:5:40;
14 for ii = m_incr
15     veh.M = M_base + ii;
16     OpenLAP_mod
17
18     LapTime = [LapTime, sim.laptime.data];
19     fuel_cons = [fuel_cons sim.fuel_cons_total.data];
20 end
21
22 M_array = M_base : 5 : (M_base + 40);
23 carburante = 0;
24 M_current = M_base;
25 n_laps = 40;
26 for ii = 1:n_laps
27     % Predictor
28     fuel_lap = interp1(M_array, fuel_cons, M_current, 'linear', 'extrap');
29
30     % Corrector
31     M_current1 = M_current + fuel_lap;
32     fuel_lap1 = interp1(M_array, fuel_cons, M_current1, 'linear', 'extrap');
33
34     carburante = carburante + fuel_lap1;
35     M_current = M_current + fuel_lap1;
36 end
37
38 % Verifica
39 for ii = 1:n_laps
40     fuel_lap = interp1(M_array, fuel_cons, M_current, 'linear', 'extrap');
41     M_current = M_current - fuel_lap;
42 end
43 end

```

The function first performs some simulations in order to extract the LapTime and the fuel consumed over a lap by varying the starting weight of the car.

In this part the Predictor-Corrector scheme is applied: since we know that the vehicle must finish the race with an empty tank ($\text{veh.M}(\text{end}) = \text{M_Base}$) the race is performed backwards from Finish to Start.

At lap ii:

1. **Predictor:** The solver first evaluate the fuel consumed assuming the car at the start of the lap to be as heavy as it is at the end.

$$\tilde{c} = f(M_{\text{end}})$$

2. **Corrector:** Then the car's weight is updated by adding the provisional fuel consumed obtained. Since the car its now heavier, its fuel consumption will be higher, so the new corrected amount of fuel consumed over the lap is extracted

$$c_{\text{true}} = f(M_{\text{end}} + \tilde{c})$$

3. **Final Update:** The car current weight is finally updated and the next iteration is performed

$$M_{\text{start}} = M_{\text{end}} + c_{\text{true}}$$

Iteration after iteration, the predictor-corrector scheme loads the car by the right amount of fuel it consumes lap after lap. Then each single fuel value is summed up to obtain the initial fuel amount to load at the start of the race

Fuel to load	Total Race Time
33.6722 kg	1h:14min:37.8sec

Table 19: Push Race

5.3.5 Constant Lift Race

The Constant Lift Race Strategy allows to load less fuel at the start of the race and so to have a lighter and faster car, at the cost of performing some Lift every lap. The L&C strategy is not always convenient, but depending on the fixed lift distance we might gain some time advantage from the strategy or completely lose time.

To investigate possible optimal solutions, three different lift strategies are employed:

- Fixed Optimal Distance in Prima Variante
- Fixed Optimal Distance in Lesmo 1
- Fixed Optimal Combined Distance in Prima Variante and Lesmo 1

For each of the two strategies a full set of simulations are performed over a set off possible lift distances, each single combination performs the full race simulation considering the right amount of fuel to be loaded with the help of the function `Calcolo_Carburante` previously explained.

The minimum Race Time is extracted and so the corresponding optimal lift distance:

Strategy	Fuel to Load [kg]	Opt. PV Lift [m]	Opt Lesmo1 Lift [m]	Gain on Push Race [sec]
PV Const Lift	33.6395	6.70	0	-0.008678
Lesmo1 Const Lift	33.6502	0	3.27	-0.008068
Combined Const Lift	33.6186	6.30	2.47	-0.017648

Table 20: Constant Lift Race

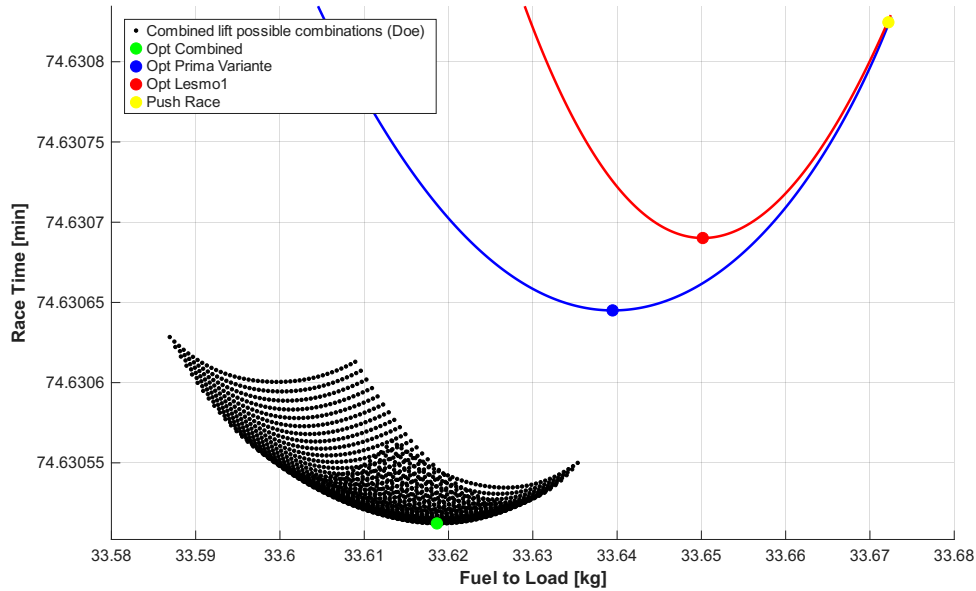


Figure 95

A first consideration can be done on the time gained in performing the L&C; no matter the strategy, the optimal gain is minimal and not sufficiently large to justify the convenience of under-loading the car. This is mainly due to the PowerTrain of the applied formula car, which is not sufficiently powerful to reach high speeds and so exploit the aerodynamic resistance.

Secondly, the data clearly demonstrates that the Combined Lift Strategy is superior to any single-corner approach. The reason lies in the non-linear physical nature of aerodynamic resistance and the consequent optimization of the **Marginal Cost (MC)**.

The Marginal Cost of Coasting, expressed in [s/kg], is defined as the inverse of the Efficiency Index:

$$MC = IE_{fuel}^{-1}$$

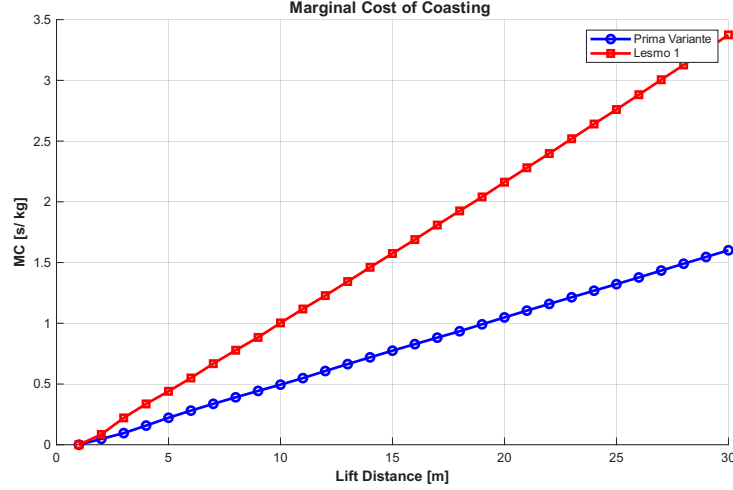


Figure 96

It quantifies the exact lap time penalty incurred to "purchase" one kilogram of saved fuel at a specific point on the track. The first few meters of lift are extremely "cheap" (low MC) because the car is at maximum speed and aerodynamic drag ($D \propto v^2$) provides massive deceleration without mechanical braking. However, extending the lift distance causes the vehicle to lose significant momentum; the aero drag diminishes, and the time penalty grows exponentially (high MC).

If the strategy relies solely on Prima Variante (PV Const Lift) to save the required amount of fuel, the algorithm is forced to extend the lift, pushing into the "expensive" region of the MC curve. The Combined Strategy resolves this limitation by distributing the cost between the two curves: it utilizes the highly efficient initial meters at Prima Variante (6.30m), and before the MC becomes too severe, picks some lift meters from the more efficient initial meters at Lesmo 1 (2.47m).

By distributing the maneuver across the two corners, the vehicle exclusively operates in the most efficient regions of the Marginal Cost curves, minimizing the overall time penalty.

5.3.6 Dynamic Lift Race

The Dynamic Lift Strategy changes the lift distance lap by lap making the vehicle adaptive to its current condition.

For racing applications where regulations don't impose a specific initial fuel load to have, an under-loaded car at the starting grid opens the possibility to adopt a totally different race strategy.

The advantage of having a lighter car is a condition that has to be maintained and maximized as long as possible during the race. From a theoretical point of view, without considering any lateral effect like tire degradation, the **L&C required to regain the fuel not initially loaded has to be performed towards the end of the race**, to conserve as much as possible the weight advantage of a lighter and faster car.

To solve this dynamic problem without incurring into huge computational costs of running the full physical solver (OpenLAP) for every single case, a sophisticated strategic algorithm is developed based on **Marginal Costs**.

1. **Lookup Tables Creation:** A broad Design of Experiments (DoE) is conducted using OpenLAP to evaluate the vehicle's Lap Time (T) and Fuel Consumption (F) across a 3D grid of input variables: Vehicle Mass (M), Lift Distance at Prima Variante (m_{PV}), and Lift Distance at Lesmo 1 (m_{Lesmo1}).

The discrete results are then interpolated to generate continuous set of results that can be extracted by the strategic algorithm.

2. Dynamic Optimization Algorithm:

The virtual strategist builds the strategy lap-by-lap using two primary metrics:

- **The Time Budget (Future Value, FV):** By starting the race with less fuel, the car is lighter and faster than the fully-fueled baseline. The time advantage gained per kilogram of reduced mass is defined as the Mass Sensitivity (S_M). As the race goes on, this advantage accumulates. The maximum acceptable time penalty (Budget) the algorithm is willing to pay at any given lap k is formulated as:

$$FV_k = S_M \cdot k \quad [\text{s/kg}]$$

- **The Marginal Cost (MC):** For any possible combination of lift distances, the algorithm computes the local cost of purchasing fuel, defined as the lap time lost divided by the fuel saved relative to the "Full Push" baseline at that specific mass:

$$MC = \frac{\Delta T}{\Delta F} \quad [\text{s/kg}]$$

At each lap k , the algorithm applies an economic constraint: it filters out all lift combinations where the marginal cost exceeds the cumulated budget ($MC > FV_k$). Among the remaining valid options ($MC \leq FV_k$), the algorithm selects the lift combination that maximizes the absolute fuel saved (ΔF).

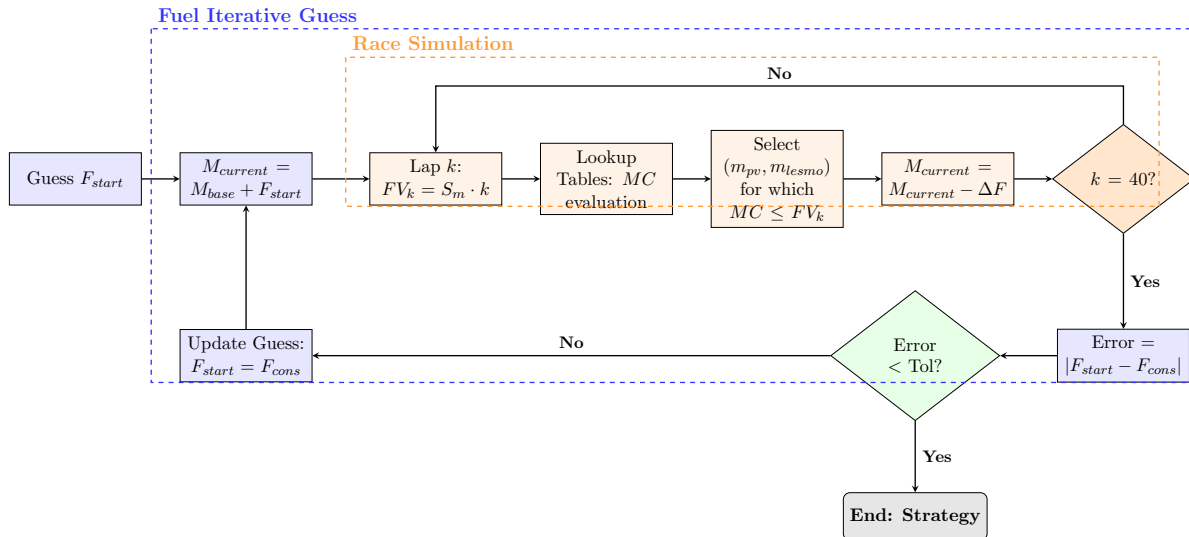


Figure 97: Dynamic Lift Algorithm Flowchart

This lap-by-lap simulation is wrapped in an external solver (Shooting Method) that iteratively adjusts the initial fuel guess until the dynamic strategy naturally

converges to an empty tank exactly at the finish line.

The following results shows the lift strategies employed for two different cases:

- Dynamic Optimal Distance in Prima Variante

Fuel to load	Gain on Push Race [s]
33.6407 kg	-0.009888

Table 21: Push Race

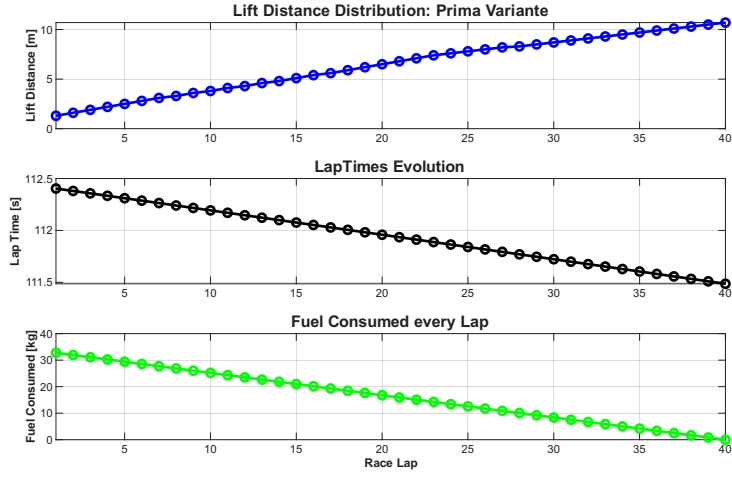


Figure 98

- Dynamic Optimal Combined Distance in Prima Variante and Lesmo 1

Fuel to load	Gain on Push Race [s]
33.6213 kg	-0.020784

Table 22: Push Race

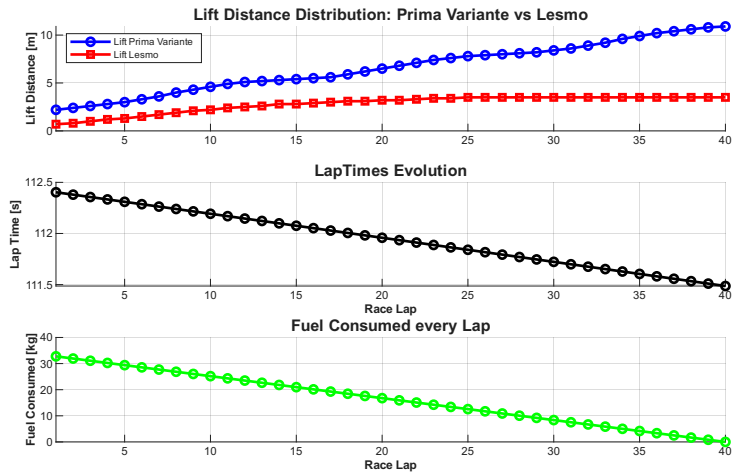
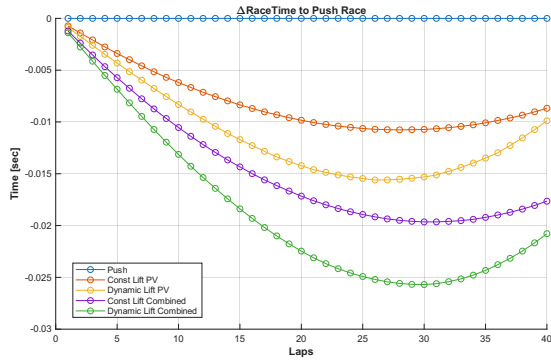
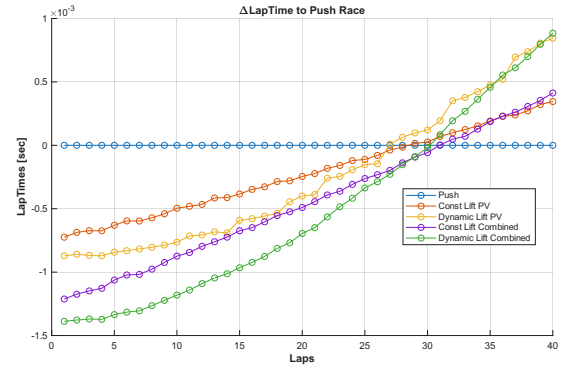


Figure 99

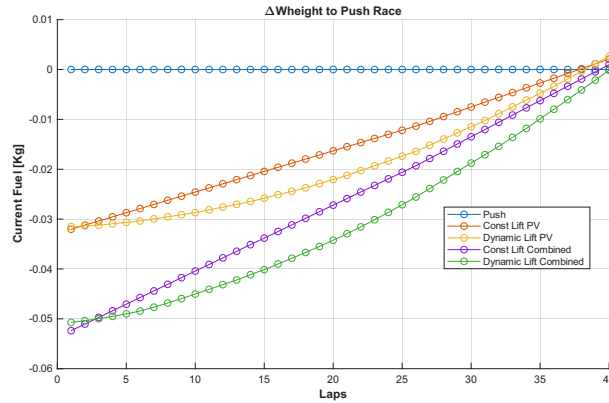
5.3.7 Final Comparison



(a) $\Delta\text{RaceTime}$



(b) $\Delta\text{LapTime}$



(c) ΔFuel

A general final comparison is done between all the race strategies. The Dynamic strategy ends to be faster on the overall race than its Constant counterpart thanks to the possibility of distributing intelligently the lift maneuver lap after lap and so exploit the lighter condition in the first part of the race.

This ability is clearly visible in the Lap Time evolution over the race: the dynamic strategies perform faster lap times at the beginning. However, they become slower than their constant counterparts towards the end of the race due to the necessity of recovering the fuel gap with huge L&C.

Combined Lift Strategies Comparison The Combined Lift Dynamic Strategy appears to be the fastest overall. if compared to the Prima Variante Dynamic Strategy, the lift sequence in Prima Variante is almost perfectly identical in both scenarios. This counter-intuitive behavior comes from the nature of the marginal cost optimizer. The solver does not operate on a fixed "quota" to split among corners, but rather on the economic budget FV_k . At any given lap, maximizes the Marginal Cost MC of both the corners. Since the aerodynamic profile of Prima Variante does not change, the algorithm pushes the lift distance in this corner to the exact same threshold dictated by FV_k , regardless of whether Lesmo 1 is being utilized or not.

However, the true supremacy of the Combined Dynamic strategy lies in unlocking a second source of fuel saving: Lesmo1. By harvesting the "cheap" initial meters of lift at Lesmo

1 the algorithm can afford to start the race with an even lower initial fuel load. This translates to a lighter, faster vehicle from the very first lap.

5.3.8 Final Comments and Model Neglection

The results demonstrate that, for the specific Tatuus vehicle configuration, the lap-time gains provided by the optimized Lift & Coast strategies are marginal compared to a pure "Push" race baseline. Furthermore, the implementation of such strategies requires extreme precision from the driver, as even minor deviations from the optimal lift-off points would likely conduct to opposite time losses.

However, this study serves a fundamental purpose: it proves how much Lift & Coast can be useful in real life applications. The potential of these strategies becomes significantly more pronounced when considering real-world race conditions, where the vehicle's performance is driven by several factors, which are currently neglected by the modified OpenLAP simulator:

- **Tyre Wear and Thermal Degradation:** Lift & Coast maneuvers reduce both longitudinal and lateral load cycles on the tyres. By alleviating mechanical stress, this strategy can extend the tyre's life, allowing the driver to maintain competitive lap times for more laps
- **Brake System Reliability:** Early throttle release reduces the kinetic energy that must be dissipated during braking. This lowers peak disc temperatures, significantly reducing brake wear
- **Engine Thermal Management:** Constant high-load operation is the primary cause of engine overheating. By reducing the time spent at Open Throttle, Lift & Coast effectively manages cooling requirements of the PowerTrain.